



RV College of Engineering®
Mysore Road, RV Vidyanikethan Post,
Bengaluru - 560059, Karnataka, India

Question Paper ID: 628695

Course Code: AI373TA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular/Supplementary Examinations February/March-2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours

Agentic Artificial Intelligence

Instructions to the students

Maximum Marks : 100

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Mention advantages of multi-agent systems over single-agent systems.	02	1	2
1.2	What are the key aspects of self-governance in agentic systems.	02	1	2
1.3	Why is proactiveness considered an important principle of agentic systems?	02	2	2
1.4	Differentiate between function calling and tool calling in LLM agents.	02	2	2
1.5	Mention two challenges of applying FF planning directly to LLM-based agents.	02	1	2
1.6	What are the essential components of interaction patterns?	02	1	2
1.7	Why is instruction formatting important in LLM-based multi-agent systems?	02	2	2
1.8	What is the role of monitoring and adaptation in agentic systems?	02	2	2
1.9	What problem does the Model Context Protocol (MCP) address in AI tooling?	02	3	2
1.10	State key difference between early generative AI tools and modern agentic systems.	02	1	1

Part B

Question No	Question	M	CO	BT
2a	Discuss the design of the Auto GPT agent system with a neat diagram.	10	1	2
2b	Define agency in the context of agentic AI systems and explain the key elements of agency.	06	2	2

3a	Explain the concept of an Intelligent Agent and discuss its core characteristics and advanced capabilities with suitable examples.	10	1	2
3b	Illustrate how an LLM responds to a query such as “What is the weather in Bengaluru?” by generating a structured tool or function call instead of directly producing an answer.	06	3	3
OR				
4a	Discuss the Key Characteristics of Reactive Architecture.	08	3	3
4b	Illustrate how tools are defined using a JSON schema format when interacting directly with LLM APIs.	08	3	3
5a	Explain Fast Forward (FF) planning as a moderately practical planning algorithm for LLM agents. Discuss its advantages, limitations, and possible adaptations for natural language planning.	10	2	2
5b	Differentiate Crew AI, Lang Chain, and AutoGen Agentic Frameworks.	06	2	2
OR				
6a	Illustrate how CWD model is used to implement the overall structure and flow of intelligent travel agent.	10	2	3
6b	Activity planning worker agent combines deep cultural knowledge with practical experience in tour operations. It excels at matching activities to traveller interests and abilities while considering factors such as seasonal availability, local customs, and logistical Constraints. Write a snippet to illustrate the Activity Planner Agent.	06	3	3
7a	Explain the Coordinator–Worker–Delegator (CWD) architecture in the context of LLM-based agentic systems.	06	1	4
7b	Illustrate Short and Long-term memory in agentic systems.	10	3	3
OR				
8a	Discuss Workflow optimization with dynamic workflows in agentic systems.	08	2	2
8b	Task Specification is useful in Intelligent Agents with a clear understanding of their duties and responsibilities. Justify your answer with an example.	08	3	4
9a	Discuss the Key capabilities of modern decision support agents.	06	2	2
9b	Discuss Model Context Protocol Server Lifecycle.	10	3	2
OR				
10a	Explain the architecture of the Model Context Protocol and describe the roles of its core components.	10	3	2
10b	Using the enterprise knowledge management use case, explain how different agents collaborate to solve complex information retrieval problems.	06	3	4